THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
FORM FOUR JOINT EXAMINATION FOR GOVERNMENT SCHOOLS –JEGOS

PRE-MOCK EXAMINATION
PHYSICS 1
MARKING SCHEME 2024

MARKING SCHEME

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
B	C	D	D	E	A	B	D	D	B

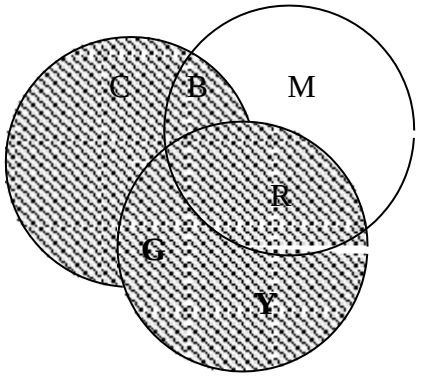
1@ = 10 Marks

2.

(i)	(ii)	(iii)	(iv)	(v)	(vi)
G	D	F	B	C	A

1@ = 06 Marks

3. (a) (i) From the following combination



C + Y = B **01**

∴ The color added is **yellow** to make a flag to be a pure green. Because **paint** absorb most of the light wavelength and reflect back only the wavelength that defines the colour you see. **02**

03 marks

(ii) That mean the real image produced on the screen by the converging lens is inverted. **02 marks**

(b) Data given

Focal length f = 5cm

Height of object h_0 = 1.5mm.

From $M = \frac{25}{f} + 1$ **00 ½**

$M = \frac{25}{5} + 1 = 6$ **01**

$M = \frac{hi}{ho}$ **00 ½**

hi = Mho.

hi = 6 x 1.5mm **01**

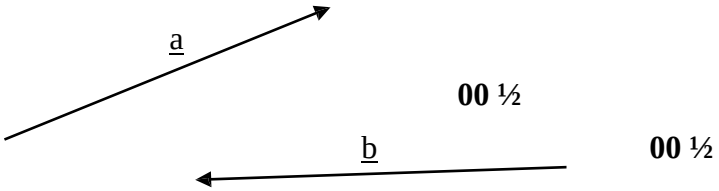
hi = 9mm

≈9cm

∴ Each division will appear to have a size of 9mm. When viewed through the simple microscope. **04 marks**

4. (a) The law state that "If two vectors are represented by two sides of a triangle in sequence, then the third closing side of the triangle drawn from laid of the first vector to the head of the second vector represented the resultant of the two vectors in both magnitude and direction. **01**

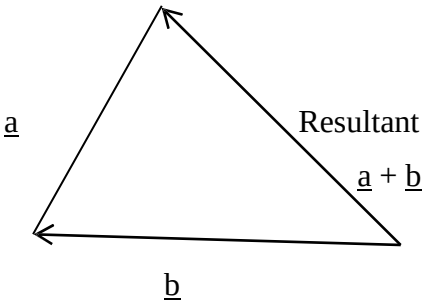
By consider two vector a and b



00 ½

00 ½

By using the law



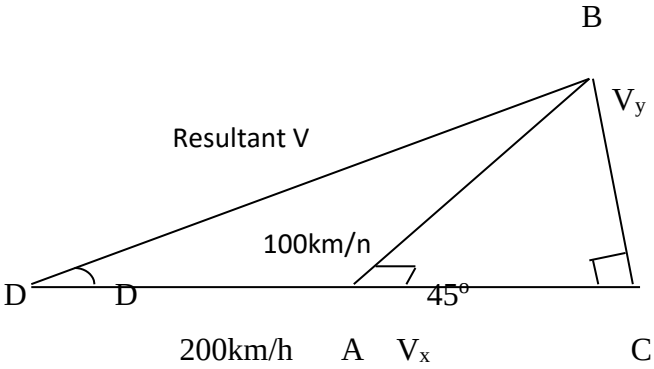
Resultant of vector **01**

a + b

The vector a and b represented the two side of triangle and vector C represent the resultant C = a + b. **01**

(b) Soln.

Diagram



01

We required to find first V_x and V_y as compound vector

Cos 45° = $\frac{V_x}{100km\ln}$ by consider ΔABC

V_x = 100 x Cos45

V_x = 70.71km/h **00 ½**

$Sin45^o = \frac{Vy}{100km/h}$

V_y = 100 x Sin45

V_y = 70.71km/h **00 ½**

The resultant velocity component (V_{x1}, V_{y1})

V_{x1} = 200km/h + 70.71km/h

V_{x1} = 270.71km/h

V_{y1} = 0+70.71

V_{y1} = 70.71km/h **00 ½**

$\therefore V2 = V_{x1}^2 + V_{y1}^2$

$V = \sqrt{(270.71)^2 + (70.71)^2}$

$V = \sqrt{73.283 + 9 + 4999.9}$

V = 279.8km/h. **01**

Direction

$SinD = \frac{V_{y1}}{V}$ **00 ½**

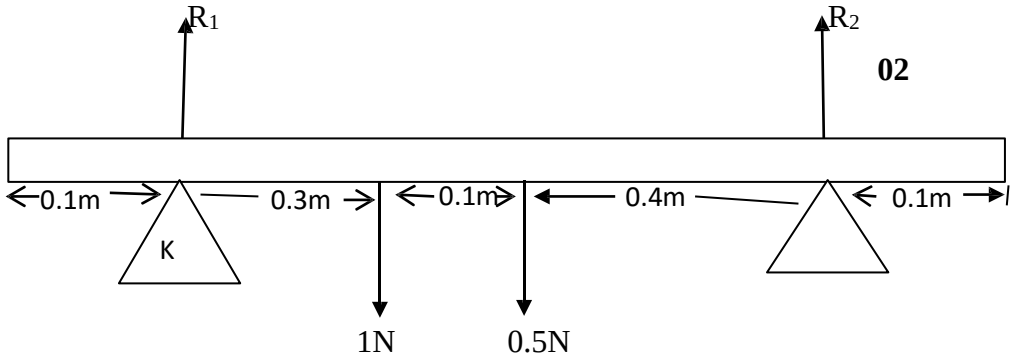
$D = Sin-1\left|\frac{(70.71)}{(279.8)}\right|$ **01**

D = 14.6°

∴ The resultant velocity is 279.8km/h at angle to 14.6° from the east.

5. (a) Momentum is conserved. The lump of plasticize increases the mass of the trolley, but has no momentum in the direction of the trolley. The momentum MV remain constant and since mass has increased the velocity reduces.**03**

(b) Diagram



Moment R_2 x 0.8 = 0.3 x 1 + 0.5 x 0.4 **01**

$$R_2 = \frac{0.3+0.2}{0.8}$$

$$R_2 = \frac{0.5}{0.8} = 0.625N$$ **01**

Force $R_1 + R_2 = 1 + 0.5$

$R_1 = 1.5 - R_2$.

$R_1 = 1.5 - 0.625$

$R_1 = 0.875N$ **01**

∴ Reaction at the support are 0.626N and 0.875N. **01**

6. (a) Advantage of anomalous exp.

- It helps in living of aquatic organism. **01 ½**

Disadvantage

- Can cause accident to ship due to submerged of ice in ocean.

(b) Data given

Initial amount = No = 400 min-1

Half life $\frac{t_1}{2}$ = 4 days

Time t = 12 days

(i) From $\frac{N}{No} = \left(\frac{1}{2}\right)^{t/\frac{t1}{2}}$ **01**

$$\frac{N}{400} = \left(\frac{1}{2}\right)$$

$$\frac{N}{400} = \left(\frac{1}{2}\right)^3 \quad \textbf{01}$$

$$N = \frac{400}{8} = 50\text{min}^{-1}$$

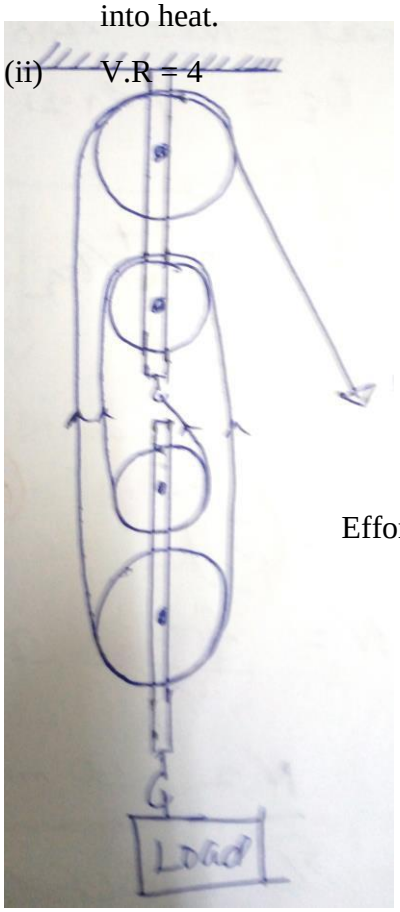
$N=50\text{min}^{-1}$. **02**

(ii) Because the breakdown of nuclei does not occur steadily but takes place at random. **02**

7. (a) (i) Because: Energy is needed to lift the bottom pulley and the ropes **01**

(ii) Friction between the moving parts of machine transform some of the energy into heat. **01**

(ii) V.R = 4



Effort **(03)**

(b) Data given

V.R =12

Total length = 2.5m.

But total length = $E_A + L_A$ where EA = Effort arm

L_A – Land arm

EA + LA = 2.5 ___(i) **01**

$$V.R = \frac{E_A}{L_A}$$

E_A = V.R x L_A

= 12 x L_A

$E_A = 12L_A$ _____ (ii) **01**

Subst. eq. (ii) int (ii)

$E_A + L_A = 2.5$

$12L_A + L_A = 2.5$

$13L_A = 2.5$

$$L_A = \frac{2.5}{13}$$

L_A = 0.19m **01**

$E_A = 12L_A$

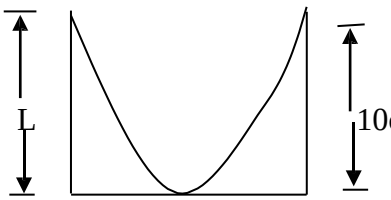
= 12 x 0.19

E_A = 2.28m ≈ 2.3.n **01**

Therefore the effort applied at 2.3m from fulcrum.

8. (a) This occur due to the internal forces in the earth's crust cause the rocks to move past each other. As the rock move they can become rocked together due to friction and their energy is stored, when the friction is overcome the rocks ship suddenly and release their stored energy in the form of wave that travels through the rocks and cause the shaking. **03**

(b) (i)



00 ½

Given data

V = 340m/s

L = 10cm,

F5=?

$l = \frac{1}{4}\lambda$

λ=4⌊

λ = 4 x 10 = 40cm

λ = 0.4m **00 ½**

$V=f\lambda \rightarrow f = \frac{V}{\lambda}$

$f_0 = \frac{V}{\lambda}$

$f_0 = \frac{340}{0.4} = 850Hz$ **01**

fn=(2n+1)fo

f_5 = (2 x 5 + 1) fo **01**

f_5 = 11fo

f_5 = 11 x 850Hz

f_5=9350Hz

∴ The frequency of the 5th overtone = 9350Hz. **01**

(ii) If temperature changes, the velocity of sound changes since velocity depend on temperature, so frequency will also change. **02**

SECTION B

9. (a) (i) Because bat produce sound once is reflected back and hence it uses echo for navigation. **02**

(ii) At the certain point, the bridge would start oscillating to the same rhythm as that of the marching steps. This oscillation would reach a maximum peak resonance when the bridge cars no longer sustain its own strength and hence collapses. **03**

(b) (i) Data given

Resistance due to primary coil = R_P = 80Ω.

Resistance due to secondary coil $R_S = 320\Omega$

From

$\frac{N_P}{N_S} = \frac{V_P}{V_S}$ i.e. V = 1R

$\frac{N_P}{N_S} = \frac{I_P R_P}{I_S R_S}$ but $\frac{I_S}{I_P} = \frac{N_p}{N_s}$ **01**

$\frac{Np.Is}{Ns\ Ip} = \frac{Rp}{Rs}$

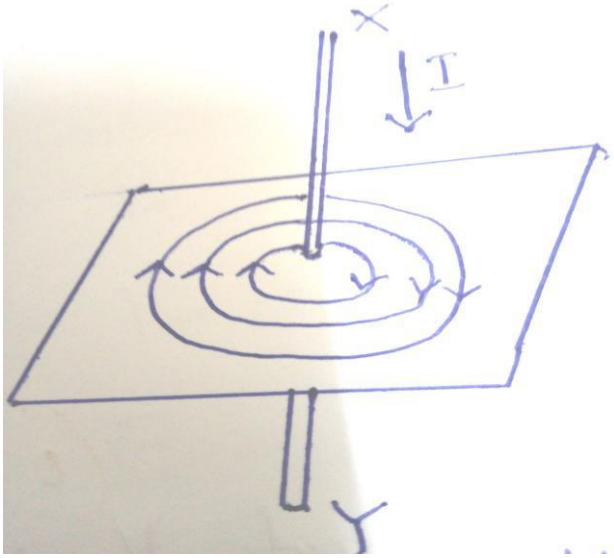
$\frac{Np}{Ns} \cdot \frac{Np}{Ns} = \frac{Rp}{Rs}$

$\left(\frac{Np}{Ns}\right)2 = \frac{Rp}{Rs} = \frac{80}{320} = \frac{1}{4}$ **01**

$\frac{Np}{Ns} = \sqrt{\frac{1}{4}} = \frac{1}{2}$ **01**

$\therefore$ The ratio is $\frac{Np}{Ns} = 1:2$ 01

(ii) As magnetic lines of force due to current in the straight conductor XY.



02

The Arrows on the magnetic liens of force shows the direction of magnetic field.

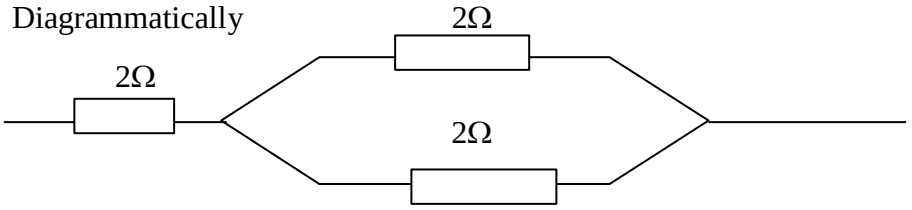
The magnitude of magnetic field at a point depends on:

- The strength of current in conductor. **02**
- The distance of point from the conductor. **02**

10. (a) (i) Short circuit faults can be identified by the fact that the circuit or that sections of circuit drawn an usually high current. Leading to fuse blowing. **02**

(ii) The circuit below will produce 3Ω resistors.

Diagrammatically



03

From equivalent resistance = $2\Omega + \frac{2\Omega \times 2\Omega}{2\Omega + 2\Omega}$

$= 2\Omega + 1\Omega = 3\Omega$

$\therefore$ The resistance 3Ω can be produce by using 2Ω.

(b) Data given

$P.d_1(V_1) = 12V$

$R_1 = 20\Omega$

$R_2 = 45\Omega$

$P.d_2(V_2) = 13.5V$

From $E = I(R+r)$ **01**

$I = \frac{E}{R+r}$ but V=1R

$$I = \frac{V}{R}$$

$I = \frac{V_1}{R_1} = \frac{E}{R+r}$ **01**

$\frac{12}{20} = \frac{E}{20+r}$

$0.6(20+r) \equiv E$

$E = 12 + 0.6r$ _____ (i) **01**

$I = \frac{E}{R_2 + r}$

$\frac{V_2}{R_2} = \frac{E}{R_2 + r}$

$E = 13.5 + 0.3r$ _____ (ii) **01**

Equate eq. (i) and eq. (ii) **02**

$12+0.6r = 13.5 + 0.3r$

$0.6r + 0.3r = 13.5 - 12$

0.3r = 1.5 **01**

$$r = \frac{1.5}{0.5}$$

r = 5Ω **02**

for E

E = 12 + 0.6 (5)

E = 12+3

E = 15V **02**

∴ The e.m.f is 15V and internal resistance is 5Ω.

11. (a) Given;

Voltage (V) = 240V

Electric power = 2.5Kw

Capacity of fuse = 15A

Current in one heater (I) = $\frac{electric\ power}{voltage}$ 01 marks

Current in one heater (I) = $\frac{2500W}{240V} = 10.4167A$

Current for the heaters (I) = $2\times 10.4167A = 20.8334$ ……. 01 $Marks$

Therefore, the fuse cannot support the operation of the two heaters. It will burn off. [02 Marks]

(b) Given

Voltage in primary coil (V_p) $= 240V$

Power in primary coil (P_p) $= 60W$

Number of turns in primary coil (N_p) $= 3000turns$

Number of turns in secondary coil (N_S) $= 200turns$

Efficiency (E) = 80%

Current in primary coil (I_p) =?

Current in secondary coil (I_s) = ?

From;

$\frac{V_p}{V_s} = \frac{N_p}{N_s}$ ……. 01 marks

$V_s = \frac{V_p N_s}{N_p}$

$V_S = \frac{240V\times 200}{3000}$

$V_S = 16V$

Electric power in primary coil = $V_p I_p$

$I_p = \frac{electric\ power\ in\ primary\ coil}{V_p}$

$I_p = \frac{60W}{240V}$

I_p = 0.25A

Therefore, current in primary coil (I_p) = 0.25A …… 01 marks

Efficiency (E) of transformer = $\frac{electric\ power\ in\ secondary\ coil}{electric\ power\ in\ primary\ coil} \times 100\%$ ……. 01 marks

Electric power in secondary coil = $\frac{electric\ power\ in\ primary}{100\%} \times efficiency$

Electric power in secondary coil = $\frac{60W}{100\%} \times 80\%$

Electric power in secondary coil = 48W …… 01 marks

Then;

From

Electric power in secondary coil = $V_s I_s$

48W = $16V \times I_s$

$I_s = \frac{48W}{16V} = 3A$

Therefore; current in secondary coil = 3A …… 01 Mark

(c) Factors for power loss in transformer;

(i) **Resistance of the coil.** ---------- [00½ Mark]

- Resistance of the coil causes power loss due to heating effect of electric current. High resistance coil causes much heating.

----------[00½Mark]

- This is minimized by using thick copper wire. Thick wires have large cross sectional area which lowers resistance. ------ [00½ Mark]

(ii) **Eddy current in the core.** ------ [00½ Mark]

- Eddy current causes heating in the core. ------ [00½ Mark]
- Eddy currents are minimized by using laminated core. - [00½ Mark]

*(iii)***Hysteresis loss**------ [00½ Mark]

- Magnetization and demagnetization of the core (**Hysteresis loss**) may cause power loss if poor magnetic material is used. [00½ Mark]
- Effect of magnetization and demagnetization can be minimized by using soft iron core. Soft iron can easily be magnetized and demagnetized. ------ [00½ Mark]

*(iv)***Leakage of field lines.** ------ [00½ Mark]

- This occurs when there is poor set-up of the coils of transformer. ----- [00½ Mark]
- Leakage of field lines is minimized by winding secondary coil on top of primary coil. ------ [00½ Mark]